Project Report: Classification of Sports and Politics Documents

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1. Introduction

In the era of information overload, the ability to automatically categorize text is essential. Whether it is for news aggregation, content filtering, or archival purposes, machine learning offers a way to handle massive datasets that would be impossible for humans to manage manually. This project explores a binary classification problem: distinguishing between **Sports** and **Politics**. By comparing three distinct machine learning techniques: Naive Bayes, Logistic Regression, and Support Vector Machines (SVM): this study identifies the most robust approach for classifying domain-specific newsgroup data.

2. Data Collection and Dataset Acquisition

For this assignment, I utilized the **20 Newsgroups dataset**, which is a collection of approximately 18,000 newsgroup posts across 20 topics.

- **Source and Fetching**: I used the fetch_20newsgroups utility from scikit-learn to programmatically download the data.
- **Filtering for Relevance**: To create a focused binary dataset, I pulled specific sub-categories:
    - **Sports**: rec.sport.baseball and rec.sport.hockey.
    - **Politics**: talk.politics.guns, talk.politics.mideast, and talk.politics.misc.
- **Data Cleaning**: It was crucial to strip out "headers," "footers," and "quotes." If I had left the headers in, the model might have cheated by learning from the newsgroup names or email addresses rather than the actual message content.

3. Dataset Analysis

After cleaning and filtering, the final dataset consisted of **4,618 documents**.

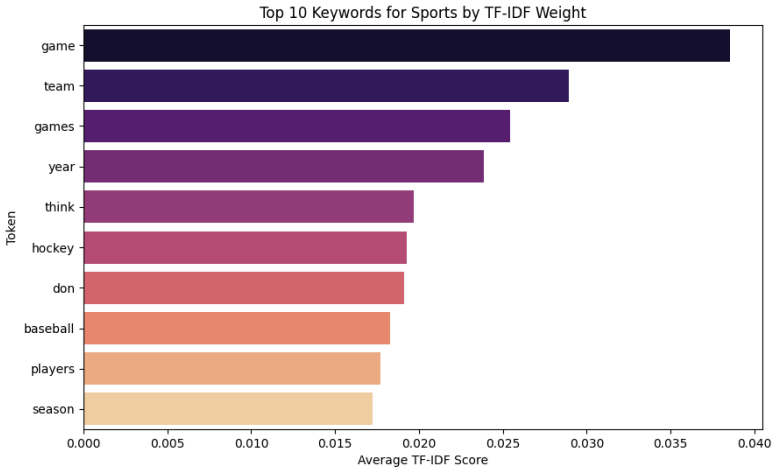
- **Class Distribution**: The dataset is relatively balanced, with **2,625 Politics** samples and **1,993 Sports** samples. This balance ensures that the models are not heavily biased toward one class.

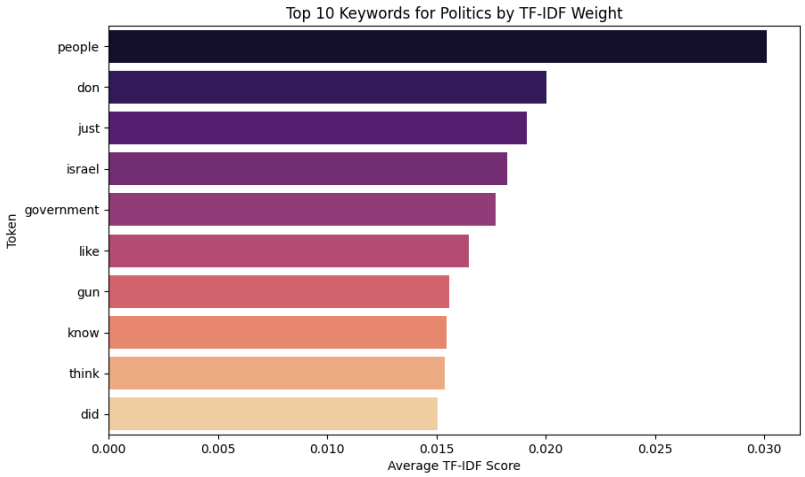
- **Vocabulary Characteristics**: One of the most interesting parts of this analysis was looking at the distinct vocabularies. Sports documents are very technical and concrete (e.g., "puck," "playoffs"), while Politics documents use more abstract or geopolitical language (e.g., "sovereignty," "amendment").

4. Feature Engineering: TF-IDF Representation

Machine learning models cannot read text directly; they need numbers. I chose **TF-IDF (Term Frequency-Inverse Document Frequency)** because it is much more effective than simple word counts.

- **The Logic**: TF-IDF weights words so that high-frequency words that appear everywhere (like "the" or "is") are ignored, while "signature words" that define a category (like "constitution" or "umpire") are given high scores.
- **Dimensionality**: I limited the vocabulary to the top **5,000 features** to prevent the model from getting bogged down by noise or rare misspellings





5. Comparative Analysis of ML Techniques

I compared three models, each with a different mathematical approach to the problem:

1. **Multinomial Naive Bayes (NB)**: A probabilistic model. It is very fast and works well even with smaller datasets. It calculates the likelihood of a category based on the product of the probabilities of each word appearing in that category.
2. **Logistic Regression**: A linear model that estimates the probability of a class. It is highly interpretable and often a go-to for text classification.
3. **Linear Support Vector Machine (SVM)**: A discriminative classifier. It tries to find the best possible line (or hyperplane) that separates the two groups with the largest possible "safety margin."

6. Quantitative Results and Comparison

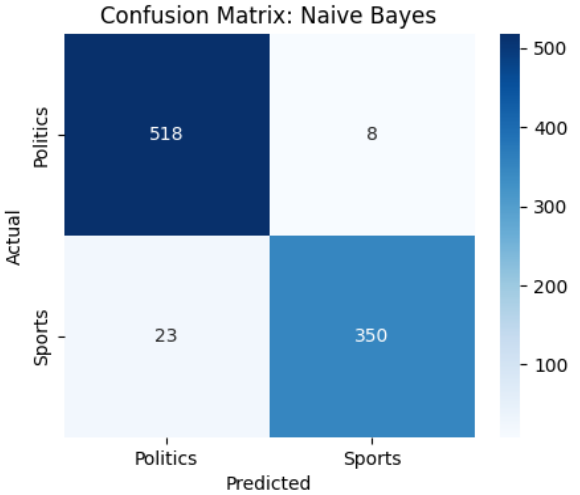
I used an 80-20 split for training and testing. The test set consisted of 899 documents.

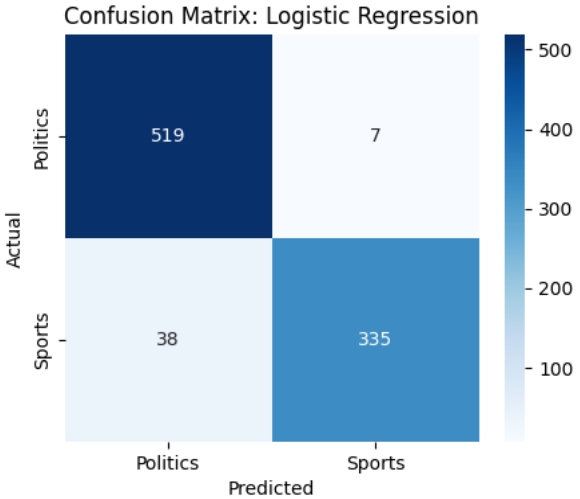
6.1 Accuracy Comparison

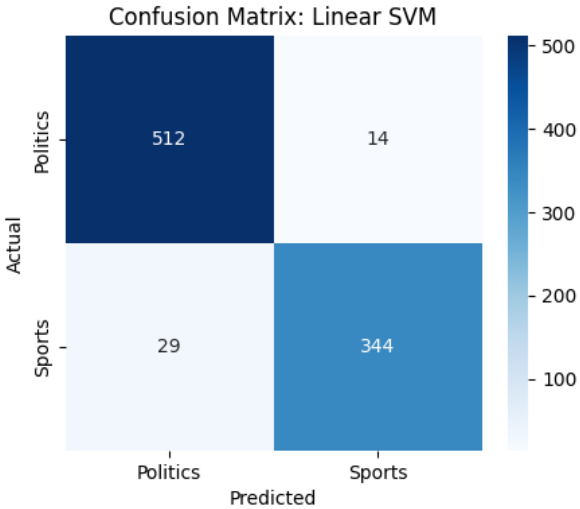
Model	Test Accuracy
Multinomial Naive Bayes	96.55%
Linear SVM	95.22%
Logistic Regression	94.99%

6.2 Detailed Breakdown

The **Naive Bayes** model performed the best. When looking at the confusion matrix, it was very interesting to see that the precision for Sports was **0.98**, but the recall was **0.94**. This means that while the model is almost never wrong when it predicts "Sports," it occasionally misses some sports documents, likely those where the language overlaps with legal or political issues (like stadium funding or sports laws).


Confusion Matrix: Naive Bayes
Actual
Politics
Sports
518
8
23
350
Predicted
Politics
Sports
500
400
300
200
100


Confusion Matrix: Logistic Regression
Actual
Politics
Sports
519
7
38
335
Predicted
Politics
Sports
500
400
300
200
100


Confusion Matrix: Linear SVM
Actual
Politics
Sports
512
14
29
344
Predicted
Politics
Sports
500
400
300
200
100

7. Limitations and Error Analysis

No system is perfect. One big limitation is the **"Bag of Words" assumption**. Because the models look at word frequency but not word order, they miss context. For example, the sentence "The athlete had a political dispute" might confuse the model because it contains strong signals for both categories. Additionally, since the data is from the 1990s, it might not handle modern slang or newer political terms effectively.

8. Conclusion

The project was a success. With **96.55% accuracy**, Naive Bayes proved to be a powerful tool for this task. The results show that even simple linear and probabilistic models can be incredibly effective if the feature engineering (TF-IDF) is done correctly.

9. Appendix: Performance Metrics

The following table summarizes the precision, recall, and f1-score for all models used in the study.

Model	Politics F1-Score	Sports F1-Score
Naive Bayes	0.97	0.96
SVM	0.96	0.94
Logistic Regression	0.96	0.94